Degree-Separated Quadratic Extraction

A source-coding improvement for explicit consistency equations

Research appendix for ProveIt

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Abstract

Separating constant, linear, and quadratic reversed kernels removes all cross-degree label constraints. A frozen certificate for the original 31-variable, 22-equation quadratic source has convolution height 2161 instead of 5556. The source identities, coefficient bounds, unconditional positivity, and optional all-real positivity repair are proved below, with exact finite verification artifacts. This appendix is a local source theorem, not an end-to-end consistency proof or a claim of a globally simplest equation.

1 Degree-separated source extraction

The gain in this section comes from changing the multiplication architecture, not from finding a smaller code for the previous homogenized square. Linear and quadratic monomials are placed in separate kernels, so that their labels may coincide. For the explicit code below the linear monomial N and the quadratic monomial J^2 both have label 2; this would be prohibited by every previous dummy-coordinate isolation condition.

1.1 Source and exact finite identity

Let $r_1, \dots, r_{22}$ be the unshifted residuals in the original 31-positive-variable package-31 source, with the complete original four-port tuple (β, K_0, M_C, M_F); M_F already includes its prescribed shift. First put

$$\rho_{14} = r_{14} + r_2, \qquad \rho_{15} = r_{15} + r_3, \qquad \rho_j = r_j \quad (j \neq 14, 15).$$

The inverse operations $r_{14} = \rho_{14} - \rho_2$ and $r_{15} = \rho_{15} - \rho_3$ prove equality of the common zero sets on every domain. There are now 89 records: 7 constants, 39 linear records, and 43 quadratic records. The support union comprises 1, 22, and 39 monomials in the respective degrees.

Write $\rho_j = \sum_\alpha a_{j,\alpha} p^\alpha$. For positive, distinct integer labels w_i, set

$$U(Z) = \sum_{i=1}^{31} p_i Z^{w_i}, \qquad e(\alpha) = \sum_i \alpha_i w_i,$$

$$c_{j,\alpha} = \begin{cases} 2a_{j,\alpha}, & |\alpha| = 0 \text{ or } 1, \\ a_{j,\alpha} \prod_i \alpha_i!, & |\alpha| = 2, \end{cases} \qquad D_d(Z) = \sum_{j,\, |\alpha| = d} c_{j,\alpha} Z^{q_j - e(\alpha)}.$$

Our packed expression is

$$G(Z, p) = D_2(Z) U(Z)^2 + D_1(Z) U(Z) + D_0(Z). \tag{1}$$

All exponents in the three kernels are nonnegative.

The finite certificate imposes two conditions, separately for $d = 0, 1, 2$. Every supported degree-d monomial must have a unique label among all monomials of that degree. Furthermore, a source term of D_d from row j can reach a marked target q_k after multiplication by a degree-d monomial only when $k = j$ and the monomials agree. There are respectively 1, 31, and 496 candidate monomials. The coefficient of a cross product in U^2 is 2 and that of a square is 1, so the factorial weighting gives

$$[Z^{q_j}] G(Z, p) = 2\rho_j(p) \qquad (1 \leq j \leq 22). \tag{2}$$

This is a universal identity in the 31 witness indeterminates and the four ports. Cross-degree collisions cause no interference: different homogeneous degrees never multiply the same kernel.

The labels, in the source variable order, are

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta),$$
$$(1, 7, 61, 36, 16, 85, 20, 67, 43, 11, 34, 13, 96, 80, 29, 50,$$
$$2, 109, 110, 112, 116, 91, 53, 55, 27, 101, 92, 9, 107, 37, 60).$$

Assign targets in the following row order:

$$(2, 22, 8, 15, 9, 11, 3, 17, 13, 18, 12, 20, 4, 21, 19, 1, 6, 10, 16, 14, 7, 5),$$
$$(28, 85, 286, 384, 593, 602, 642, 731, 906, 966, 1083, 1212, 1231,$$
$$1393, 1461, 1471, 1678, 1719, 1732, 1880, 2107, 2129).$$

The exact sparse kernels are supplied as machine-readable polynomials in `split_kernels.json`, derived from the canonical `split_code.json`. The checker independently parses the original residuals, proves the two inverse row identities, enumerates 22 544 source/expansion products, and compares all 22 formal output polynomials. There are exactly 89 target contributions. It obtains

$$M = 116, \quad Q = 2129, \qquad (\deg D_0, \deg D_1, \deg D_2) = (1719, 2027, 1929),$$
$$H = \max_{0 \le d \le 2} (\deg D_d + dM) = 2161. \tag{3}$$

Every kernel has internally distinct exponents. The leading term of D_2 is exactly $+Z^{1929}$, arising from the EX term in row 5.

1.2 Coefficient bounds and unconditional positivity

The complete native tuple has strict port bounds

$$\beta < 2^{1064921}, \quad K_0 < 2^{1070004}, \quad M_C < 2^{1064921}, \quad M_F < 2^{1064922}.$$

Taking absolute integer coefficients and replacing the port powers by their bit bounds yields, for every weighted kernel coefficient, absolute value strictly below

$$C = 2^{2134927}.$$

The checker recomputes this finite integer majorant from every coefficient. The common selected source floor $\beta_S = 2^{2134940}$ satisfies $\beta_S > 4096C$, so $B = \beta_S b^2 > 4096Cb^2$. For the Jones choice $B = b^3$, it suffices to retain the existing requirement $b \ge 4096C$.

Put $\ell = \sum_i B^{w_i}$ and $U = \ell + g$. If $g \ge 0$, then $U \ge B^{116}$. For $B \ge 4096C$, the certified leading coefficient gives

$$D_2(B) > B^{1929} \left(1 - \frac{C}{B-1}\right) > \frac{1}{2} B^{1929},$$

whereas $|D_1(B)| < 2CB^{2027}$ and $|D_0(B)| < 2CB^{1719}$. After division by $B^{1929}U^2$, the two potentially negative terms are less than $2C/B^{18} < 1/8$ and $2C/B^{442} < 1/8$. Thus

$$G(B, U) > \frac{1}{4} B^{1929} U^2 > 0 \qquad (g \ge 0). \tag{4}$$

This proof precedes digit recovery and does not assume that B is already a power of two. In particular, it does not incorrectly infer positivity from the signs of all kernel coefficients.

1.3 Sparse padding and the source interval

Suppose $b \ge 4$, $B \ge 4096Cb^2$, and $0 \le g < 2bB^{116}$. Distinct positive labels imply $\ell < 2B^{116}$, whence $U < 3bB^{116}$. By the actual kernel degrees,

$$|G(B, U)| < 2C(1 + 3b + 9b^2)B^H \le 26Cb^2 B^H, \qquad 2G(B, U) < \frac{1}{64} B^{H+1}.$$

Define

$$T_{\mathrm{arg}} = \sum_j B^{q_j}, \qquad N_0 = B^{117}, \qquad N_1 = B^{2162}, \qquad J_{\mathrm{code}} = 2G(B,U) + T_{\mathrm{arg}}.$$

Since $Q = 2129$, both T_{arg} and $(B-2)T_{\mathrm{arg}}$ are less than $N_1/64$. Therefore

$$0 < J_{\mathrm{code}} < N_1/32, \qquad J_{\mathrm{code}} + (B-2)T_{\mathrm{arg}} + 2 < N_1.$$

If the usual low words of S and T are each below N_0, the latter strict inequality proves $S+T < N = N_0N_1$ including their possible low-block carry. All these inequalities hold before binary recovery. The older dense padding $2G + B\sum_{h=0}^{H} B^h$ also lies between 0 and $2B^{H+1}$, if an adapter retains $N_1 = 4B^{H+1}$.

After digit recovery write $g = \sum_i n_i B^{w_i}$ with $0 \le n_i < b$ and $p_i = n_i + 1$. Then $U = \sum_i p_i B^{w_i}$ and $\sum_i p_i \le 31b$. Every coefficient of the complete polynomial G is bounded in absolute value by

$$C\left((\sum_i p_i)^2 + \sum_i p_i + 1\right) \le 993Cb^2 < B/4. \tag{5}$$

This controls unsupported collisions too; it sums the three homogeneous layers rather than assuming that all quadratic labels are distinct. The inherited sparse-borrow padding lemma now applies to $J_{\mathrm{code}} = 2G + T_{\mathrm{arg}}$. Its marked coefficients are exactly $2\rho_j$, by (2), and the row inverse recovers the original source equations.

Conversely, any positive source solution supplies $n_i = p_i - 1$ for every sufficiently large base. The all-one tuple is not a solution because the unchanged row 5 equals 1 there. Hence some $n_i > 0$, and the positive labels give $g \ge B > b$. Thus the lower source-witness bound is preserved, not merely the upper bound.

1.4 One-cell positivity repair for signed adapters

An adapter that must prove positivity for every integer g, rather than only $g \ge 0$, may use

$$G_+(B,U) = G(B,U) + B^{2162}.$$

This changes no target coefficient since $2162 > Q$, and increases the exact height only to 2162. Completing the square, and using the bounds above,

$$\begin{aligned} G_+(B,U) &\ge B^{2162} - |D_0(B)| - \frac{D_1(B)^2}{4D_2(B)} \\ &> B^{2162} - 2CB^{1719} - 2C^2B^{2125} > 0. \end{aligned}$$

Indeed the normalized subtractions $2C/B^{443}$ and $2C^2/B^{37}$ are each less than $1/4$ for $B \ge 4096C$. Positivity thus holds for every real U. The added off-target coefficient is 1, within the same coefficient bound; a sparse bounded-input adapter can take $N_1 = B^{2163}$.

1.5 What is and is not certified

The exact standard-library replay and ten corruption controls are in `verify_split.py` and `test_split.py`; ordinary and optimized Python produce byte-identical verification receipts. Neither search code nor a third-party symbolic package is a replay dependency. Compared with the frozen wave7 source, the convolution height falls from 5556 to 2161 (about 61.1%) and $H + M + 2$ falls from 5741 to 2279 (about 60.3%). These are source exponents, not final equation degrees. The finalizer must still be composed and its actual arithmetic circuit independently audited. No global optimum or new end-to-end proof-assistant certificate is claimed. The unbounded compiler/refutation and arithmetic finalizer theorems remain identified inherited semantic dependencies.